

Supplementary Information: Environmental recurrence and directional cache information select a costly memory-use allele without a resolved one-bit population benefit

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S1. Status and scope

This supplement documents two prospective studies. Part A (Sections S1-S15) documents Study 001; its content, values and provenance are unchanged from version 1.0 apart from this introduction and cross-references to Part B. Part B (Sections S16-S26) documents Study 002, a post-v1.0 prospective extension added in version 1.1.0.

Part A. Study 001 (PHASE2-MUTABLE-MEMORY-001)

Part A documents the frozen design, analysis and audit of the prospective study PHASE2-MUTABLE-MEMORY-001 (specification revision 1). The accepted primary decision is **SUPPORTS RECURRENCE-ATTRIBUTABLE SELECTIVE ENRICHMENT**; the secondary classifications are **BOUNDED BELOW THE POSITIVE ONE-BIT SCALE** for P_{abs} and **UNRESOLVED** for P_{rec} . Every value below is read from the saved analyzer, diagnostics and audit records listed in Section S15. The complete scope is 41,600 independent paired blocks, eight cells per block, 332,800 paths, 85,196,800 path-updates and 10,905,190,400 objective queries.

S2. State and full update order

Fixed state: 32 haploid individuals in slots 0-31; one unsigned 32-bit genotype per individual; one policy allele, M or F, per individual; and, for M, a private cache that is invalid or holds one unsigned 32-bit genotype. F individuals hold no valid cache. Lineage identifiers are audit metadata and are never read by scientific operators.

Initialization (update 0) for each block: 32 independent uniform 32-bit genotypes shared by all eight cells; all caches invalid; every label F in ALL_F cells and every label M in ALL_M cells.

At each update $t = 1, \dots, 256$ the following steps occur in this order:

1. Generate the target innovation I_t and recurrence bit R_t and set the target T_t according to the cell's law (Section S3).
2. For each parent slot $i = 0, \dots, 31$, construct four candidates: parent, policy probe, global scout and local child (Section S4). All masks, local-bit draws and donor keys are generated for every coordinate whether used or not.
3. Evaluate all 128 candidates by Hamming mismatch to T_t . Candidate order is (family, parent slot), with family order parent, policy probe, global scout, local child.
4. Draw 32 survivors sequentially without replacement with exact integer weights $2^{(32-h)}$ (Section S5). Survivor slots are assigned in draw order.
5. Each survivor inherits its producing parent's policy; then each survivor slot independently flips its policy with probability $1/32$ (Section S6).

6. Assign caches: post-mutation F survivors receive an invalid cache; post-mutation M survivors receive their producing parent's pre-update genotype.

7. Record memory-use count, total population mismatch, valid cache-probe use, cache-probe survivors, F -> M and M -> F counts and the query count, all after step 6.

True population accuracy at update t is $1 - (\text{mean survivor mismatch})/32$. The late window is updates 193-256 inclusive.

S3. Target laws

For every block and update, an indexed uniform innovation I_t and an indexed fair recurrence bit R_t are generated whether used or not.

- ZERO: $T_t = I_t$ for all t .
- HALF: $T_1 = I_1, T_2 = I_2$; for $t \geq 3$, $T_t = T_{(t-2)}$ if $R_t = 1$ and $T_t = I_t$ otherwise.

ZERO and HALF use the same innovation at every non-recurrent update. HALF is a prior, outcome-informed choice that exactly matches the one-generation cache delay to a two-update target return, and therefore favors retrieval by construction. No other lag, copy probability or target law was tested.

S4. Candidate construction and the fixed budget

For parent slot i with genotype x_i at update t :

1. **Parent:** x_i .
2. **Policy probe:** for F, x_i XOR fresh_mask(t, i) with an independent uniform 32-bit mask; for a valid M in ACTIVE, the cached genotype; for an invalid M in ACTIVE, the same fresh construction as F; for any label in SHAM, the same fresh construction as F.
3. **Global scout:** x_i XOR scout_mask(t, i) with an independent uniform 32-bit mask.
4. **Local child:** the lowest-mismatch member of candidates 1-3 for that parent, ties broken by an independent keyed 64-bit donor key (lower wins) and then by family index, XOR a local mask in which each of the 32 bits flips independently with probability $1/32$.

Every parent therefore makes exactly four evaluated candidates and every update exactly 128 objective queries. Because XOR with a uniform word yields a uniform word, the fresh probe and the scout are each effectively uniform proposals. A valid ACTIVE-M carrier replaces exactly one of these two uniform proposals with its cache; this replacement is the entire cost of memory use. There is no score penalty, no tunable cost coefficient and no deduplication: a cache equal to the unchanged parent is evaluated as a separate candidate. The magnitude of the cost is conditional on the frozen fresh-probe distribution and is not a general cost scale. An invalid-cache M carrier, which can occur only at update 1, uses exactly the same fresh draw as F and SHAM, so ACTIVE and SHAM first diverge only at valid-M slots.

S5. Survival by exact integer-weighted sequential sampling

Candidate j with raw mismatch h_j receives weight $w_j = 2^{(32-h_j)}$, an exact integer proportional to $2^{(-h_j)}$. Thirty-two candidates are selected sequentially without replacement. At survivor draw $d = 0, \dots, 31$:

1. Retain unselected candidates in candidate-index order and sum their weights to W_d .
2. Obtain an unbiased integer Z_d uniform on $[0, W_d)$ from the keyed 64-bit survival stream by multiply-high rejection: for retry $r = 0, 1, \dots$, read a 64-bit word x , form the 128-bit product $m = x * W_d$, set threshold_d = $(2^{64} - W_d) \bmod W_d$, reject if low64(m) < threshold_d, and otherwise set $Z_d =$

high64(m).

3. Select the first remaining candidate whose cumulative weight exceeds Z_d .

The maximum weight sum is $128 * 2^{32} = 2^{39}$, so unsigned 64-bit totals and unsigned 128-bit products suffice. The procedure is an exact Plackett-Luce sample without replacement using no floating-point arithmetic. No label, cache, lineage identifier or penalized score enters it.

S6. Policy mutation and cache inheritance and use

Each selected candidate initially inherits the policy of the parent that produced it. Then, independently for every survivor slot, the policy flips $M \leftrightarrow F$ with probability $\mu = 1/32$. After mutation, a post-mutation F survivor's cache is invalid and a post-mutation M survivor's cache becomes the producing parent's pre-update genotype. Consequences:

- from update 2 onward every M carrier holds a valid one-step lineage cache;
- a new $F \rightarrow M$ mutant first uses its record on the next update;
- an $M \rightarrow F \rightarrow M$ history carries no older hidden record;
- the producing parent was itself a survivor of selection against the previous target, so a cache used at update $t+1$ holds a genotype that survived selection against $T_{(t-1)}$.

Under ACTIVE, the cache is read only to construct the policy probe of a valid M carrier. Under SHAM it is never read.

The allele therefore jointly determines whether the one-step record is retained and whether it is retrieved. Every individual has a cache slot, but only M carriers receive a valid record, so this design cannot separate selection on retention from selection on retrieval.

S7. Coordinate-keyed random numbers

All random values come from canonical Random123 Philox4x32-10 with multipliers D2511F53 and CD9E8D57 and Weyl constants 9E3779B9 and BB67AE85. For each purpose, the two 32-bit key words are bytes 0-3 and 4-7 (little-endian) of SHA-256 of the UTF-8 text PHASE2-MUTABLE-MEMORY-001|production-r1|<purpose>. The four little-endian 32-bit counter words are exactly (block, update, entity, subindex). Block runs over 0-41599 and update is 0 at initialization and 1-256 otherwise. Output words 2-3 are unused.

Purpose	Entity	Subindex	Extraction
INITIAL_GENO TYPE	parent slot	0	word 0
TARGET_INNO VATION	0	0	word 0
TARGET_COPY	0	0	word 0 bit 0
FRESH_MASK	parent slot	0	word 0
SCOUT_MASK	parent slot	0	word 0
LOCAL_BIT	parent slot	bit 0-31	flip iff word0 & 31 == 0
DONOR_KEY	3*parent_slot + family_index (family 0-2)	0	word0 OR (word1 << 32); lower wins

Purpose	Entity	Subindex	Extraction
SURVIVAL_UNI FORM	survivor draw 0-31	rejection retry 0, 1, ...	word0 OR (word1 <= 32)
POLICY_MUTA TION	survivor slot 0-31	0	flip iff word0 & 31 == 0

The implementation aborts if any coordinate leaves its range or a survival retry exceeds the 32-bit subindex. A mechanical enumeration of the declared coordinate schemas and a purpose-separated collision audit passed before any population path. Because every draw is addressed by coordinates and generated regardless of branch, arm, law or label, all eight cells of a block share their random values wherever the coordinates coincide.

S8. SHAM identities and the neutral derivation

In SHAM the policy probe is always the fresh construction, so the genotype process never reads a label or cache. With the same keyed draws:

- **N1.** SHAM genotype, mismatch, accuracy and survivor trajectories are bit-identical between ALL_F and ALL_M.
- **N2.** Policy labels in the two SHAM starts are exact complements after every update, because both starts share the genealogy and the same mutation draws and differ only in their initial labels. Their memory-use frequencies sum to 1 at every update, and their start-averaged late-window frequency is exactly 1/2 in every block.

Derivation of the neutral expectation: because the genealogy is independent of labels in SHAM, the label of a survivor at update t equals its founding ancestor's label passed through t independent flips, each with probability μ . With $r = 1 - 2\mu = 15/16$, the probability of an odd number of flips is $(1 - r^t)/2$. Hence $P(M \text{ at } t \mid \text{ALL_F}) = [1 - r^t]/2$ and $P(M \text{ at } t \mid \text{ALL_M}) = [1 + r^t]/2$, and the expected ALL_M minus ALL_F difference is r^t . At update 256 this is $r^{256} = 6.678005283720385e-08$. Averaged over updates 193-256 it is $(1/64) \sum_{t=193}^{256} r^t = r^{193}(1 - r^{64})/(64(1 - r)) = 9.57983820600146e-07$. These expectations justify the late window for SHAM; they do not establish convergence of ACTIVE paths.

Observed descriptive consistency: SHAM memory-use frequencies were 0.500083 and 0.499917 under ZERO and 0.499089 and 0.500911 under HALF, and the exact saved fractions in each pair sum to 1. SHAM accuracy was 0.607310 under ZERO and 0.619018 under HALF from both starts, with identical exact fractions. The record verifier confirmed both identities block by block in all 41,600 blocks.

What SHAM does not control: SHAM is an exact neutral benchmark for the label process (N1, N2; expected start-averaged frequency 1/2) and a paired control for query budget and keyed random draws, but ACTIVE genotypes and genealogies diverge from SHAM at the first valid-M retrieval, so SHAM does not hold realised drift or genotype state fixed for ACTIVE paths and ACTIVE-SHAM contrasts are arm-level comparisons within the model, not path-level counterfactuals. It also does not separate the lag-two information in the cache from the loss of one fresh proposal (the ZERO law provides that comparison through C_rec, and Study 002 provides a within-law comparison; Section S16); it does not represent alternative caches, costs or retrieval rules; and it says nothing about ACTIVE convergence or long-run behavior.

S9. Estimands and interval construction

For block b , let $L_b(\text{arm}, \text{law}, \text{start})$ be the mean memory-use frequency over the 64 late-window updates and $\text{Acc}_b(\text{arm}, \text{law}, \text{start})$ the corresponding mean accuracy. Block variables:

- $A_{\text{HALF},b} = [L_b(\text{ACTIVE}, \text{HALF}, \text{ALL}_F) + L_b(\text{ACTIVE}, \text{HALF}, \text{ALL}_M)]/2$, and $A_{\text{ZERO},b}$ analogously;
- $C_{\text{abs},b} = A_{\text{HALF},b} - 1/2$ (range length 1);
- $C_{\text{rec},b} = A_{\text{HALF},b} - A_{\text{ZERO},b}$ (range length 2);
- $D_{\text{HALF},b} = L_b(\text{ACTIVE}, \text{HALF}, \text{ALL}_M) - L_b(\text{ACTIVE}, \text{HALF}, \text{ALL}_F)$ (range length 2);
- $D_{\text{ZERO},b} = L_b(\text{ACTIVE}, \text{ZERO}, \text{ALL}_M) - L_b(\text{ACTIVE}, \text{ZERO}, \text{ALL}_F)$ (range length 2);
- with $P_b(\text{arm}, \text{law})$ the start-averaged accuracy, $P_{\text{abs},b} = P_b(\text{ACTIVE}, \text{HALF}) - P_b(\text{SHAM}, \text{HALF})$ (range length 2);
- $P_{\text{rec},b} = [P_b(\text{ACTIVE}, \text{HALF}) - P_b(\text{SHAM}, \text{HALF})] - [P_b(\text{ACTIVE}, \text{ZERO}) - P_b(\text{SHAM}, \text{ZERO})]$ (range length 4).

Each estimate is the mean of its block variable over $n = 41,600$ independent blocks. Because each estimand is linear in cell means, each saved estimate equals the same linear combination of the saved descriptive cell means; the statistics-checking script verifies these identities exactly in rational arithmetic.

Hoeffding's inequality for the mean of n independent variables each confined to an interval of length R gives $\Pr(|\text{mean} - \text{expectation}| \geq h) \leq 2 \exp(-2 n h^2 / R^2)$. Setting the right side to α_{each} gives $h = R \sqrt{\ln(2/\alpha_{\text{each}})/(2n)}$. The primary family of four estimands uses $\alpha_{\text{each}} = 0.05/4 = 0.0125$, giving 95% familywise coverage by Bonferroni's inequality; the secondary family of two uses $\alpha_{\text{each}} = 0.05/2 = 0.025$. The resulting half-widths are 0.00781023 (C_{abs}), 0.01562046 (C_{rec} , D_{HALF} , D_{ZERO}), 0.01451463 (P_{abs}) and 0.02902925 (P_{rec}). The primary range-2 half-width lies below $\Delta/2 = 0.015625$, so the gate can be passed only if a start contrast is estimated near zero. Each interval is $[\text{estimate} - h, \text{estimate} + h]$, computed from the exact rational estimate and a high-precision half-width. No empirical variance, normal approximation, within-path independence, pilot or favorable pairing is used.

S10. Frozen decision rules and their application

Primary rules, applied in order with $\Delta = 1/32 = 0.03125$:

1. Failure of a deterministic identity, seed or counter audit, saved-output reconstruction or 128-query count: **INVALID**. Not triggered.
2. Interval for D_{HALF} or D_{ZERO} not wholly inside $(-\Delta, +\Delta)$: **START-DEPENDENT; SCIENTIFIC QUESTION UNRESOLVED**. Not triggered: D_{HALF} $[-0.014898, 0.016343]$ and D_{ZERO} $[-0.015717, 0.015524]$ both lie inside.
3. Lower bounds of both C_{abs} and C_{rec} above Δ : **SUPPORTS RECURRENCE-ATTRIBUTABLE SELECTIVE ENRICHMENT**. Triggered: C_{abs} lower bound 0.302482 and C_{rec} lower bound 0.426015.
4. Lower bound of C_{abs} above Δ and upper bound of C_{rec} at or below Δ : **SELECTIVE ENRICHMENT NOT ATTRIBUTABLE TO RECURRENCE AT THE FIXED SCALE**. Not reached.
5. Upper bound of C_{abs} at or below Δ : **BOUNDED NEGATIVE AT THE ONE-INDIVIDUAL SCALE**, with a statement of whether the upper bound is below $-\Delta$. Not reached; the adverse-selection field is therefore null in the decision record.
6. Every other pattern: **UNRESOLVED**. Not reached.

The decision record lists decision rule 3 as applied. Secondary rules, applied in order to each of P_abs and P_rec with Delta_P = 1/32: lower bound above Delta_P, meaningful positive performance; upper bound below -Delta_P, meaningful adverse performance; upper bound at or below Delta_P, **BOUNDED BELOW THE POSITIVE ONE-BIT SCALE**; otherwise **UNRESOLVED**. P_abs has upper bound 0.023892, at or below 0.03125, and is BOUNDED BELOW THE POSITIVE ONE-BIT SCALE. P_rec has lower bound -0.017669 and upper bound 0.040390 and is UNRESOLVED. Secondary classifications cannot alter the primary decision. No replicate, horizon, target-law, mutation-rate or cost change follows any branch.

S11. All 22 saved estimate records

Inferential records carry frozen bounds; descriptive records do not. Exact values are the saved rational estimates; decimal values are rounded to six places.

#	Record	Exact	Estimate	Interval
1	C_abs	5287181/17039360	0.310292	[0.302482, 0.318102]
2	C_rec	75251823/170393600	0.441635	[0.426015, 0.457256]
3	D_HALF	30793/42598400	0.000723	[-0.014898, 0.016343]
4	D_ZERO	-8213/85196800	-0.00009 6	[-0.015717, 0.015524]
5	P_abs	51129277/5452595200	0.009377	[-0.005138, 0.023892]
6	P_rec	61943359/5452595200	0.011360	[-0.017669, 0.040390]
7	Frequency, ACTIVE, ZERO, ALL_F	314125/851968	0.368705	descriptive
8	Frequency, ACTIVE, ZERO, ALL_M	31404287/85196800	0.368609	descriptive
9	Frequency, ACTIVE, HALF, ALL_F	8625439/10649600	0.809931	descriptive
10	Frequency, ACTIVE, HALF, ALL_M	34532549/42598400	0.810654	descriptive
11	Frequency, SHAM, ZERO, ALL_F	42605507/85196800	0.500083	descriptive
12	Frequency, SHAM, ZERO, ALL_M	42591293/85196800	0.499917	descriptive
13	Frequency, SHAM, HALF, ALL_F	42520779/85196800	0.499089	descriptive
14	Frequency, SHAM, HALF, ALL_M	42676021/85196800	0.500911	descriptive
15	Accuracy, ACTIVE, ZERO, ALL_F	1650292613/2726297600	0.605324	descriptive
16	Accuracy, ACTIVE, ZERO, ALL_M	1650309069/2726297600	0.605330	descriptive
17	Accuracy, ACTIVE, HALF, ALL_F	1713218809/2726297600	0.628405	descriptive
18	Accuracy, ACTIVE, HALF, ALL_M	428290869/681574400	0.628385	descriptive
19	Accuracy, SHAM, ZERO, ALL_F	827853941/1363148800	0.607310	descriptive
20	Accuracy, SHAM, ZERO, ALL_M	827853941/1363148800	0.607310	descriptive
21	Accuracy, SHAM, HALF, ALL_F	210953313/340787200	0.619018	descriptive
22	Accuracy, SHAM, HALF, ALL_M	210953313/340787200	0.619018	descriptive

Derived start averages used in the main text: ACTIVE frequency 0.810292 under HALF and 0.368657 under ZERO; ACTIVE accuracy 0.628395 under HALF and 0.605327 under ZERO. The ZERO ACTIVE minus SHAM accuracy contrast, -0.001983, is descriptive. The analyzer's descriptive relation labels are DISAGREEMENT for P_abs and UNCERTAINTY for P_rec.

S12. Descriptive diagnostics

Totals over all blocks and updates, from the descriptive diagnostics record. These counts support interpretation only and carry no inferential bounds. Cache-probe use counts valid ACTIVE-M retrievals; cache-probe survivors counts retrieved candidates that were selected.

Cell	F -> M	M -> F	Cache-probe uses	Cache-probe survivors
ACTIVE, ZERO, ALL_F	7,009,928	3,640,445	119,306,059	23,380,466
ACTIVE, ZERO, ALL_M	6,382,195	4,268,178	138,178,075	27,148,189
ACTIVE, HALF, ALL_F	2,226,924	8,423,449	262,246,633	76,047,092
ACTIVE, HALF, ALL_M	1,743,217	8,907,156	276,722,365	79,823,372
SHAM, ZERO, ALL_F	5,656,667	4,993,706	0	0
SHAM, ZERO, ALL_M	4,993,706	5,656,667	0	0
SHAM, HALF, ALL_F	5,659,540	4,990,833	0	0
SHAM, HALF, ALL_M	4,990,833	5,659,540	0	0

The SHAM mutation counts are exchanged between starts, as label complementarity (N2) implies. Under HALF, ACTIVE cache probes were used more often and survived more often than under ZERO. The same diagnostics record (SHA-256 in Section S15) also archives fixation, extinction and late-window occupancy counters; they are not interpreted here. The diagnostics record lists 10,905,190,400 total objective queries.

S13. Seeds, fixtures, outputs and audit structure

Cohort. One production namespace (production-r1) and 41,600 blocks; no prior genotype, preparation, target, trajectory, random tape or outcome was reused. Production ran as one allocation (job 53530597), followed by the frozen analyzer (job 53530688) and the separately implemented internal audit job (53530689). The prospective timing of the specification, design review and production submission is given in Section S15.

Deterministic fixtures (all passed before production; none contains a sampled scientific outcome). (1) One- and two-update hand trace covering all four candidate families, donor choice, exact sequential weighted survival, policy mutation and cache transitions; (2) memory-better case (lag-two target, cache mismatch 0, fresh mismatch 16); (3) memory-worse case (fresh target, cache mismatch 28, fresh mismatch 4); (4) duplicate case with the cache equal to the unchanged candidate and 128 queries; (5) invalid-cache case in which ACTIVE-M uses the same fresh draw as F and SHAM; (6) full 256-update SHAM paths from ALL_F, ALL_M and a mixed assignment verifying N1 and N2; (7) target-law pairing at innovations, recurrent updates and the $t = 1/2$ boundary; (8) an exact weighted-selection fixture with known weights and draws, mismatches 0 and 32, a forced rejection and retry, duplicate genotypes and exact survivor order;

(9) random-coordinate invariance under changes to labels, arm, law and start; (10) regeneration of a non-audit path from its block identifier and source and configuration hashes to bit-identical summaries.

Saved outputs. Frozen specification, source, build environment, configuration and SHA-256 manifests; Philox implementation and version, key schema, namespace and collision receipt; per-update binary records for every path; per-path late summaries and final-state hashes; per-block primary and performance variables and N1/N2 flags; full candidate rows for audit blocks 0-63 (512 paths, 16,777,216 rows) including genotype, family, producing parent, labels, cache, mismatch, integer weight, selection draw, retry and integer, selected rank, target and recurrence indicator; all 22 estimate records with bounds and classifications; and job request, accounting, logs, exit status and delivery manifests.

Replay audit. The C++ tool MMEM-AUDIT-REPLAY-1.0.1 (executable SHA-256 8cf650a89e690af7e65629bee155ea08b9b985c4d504e70faf75875e1ebddc6e), run in production mode with production-namespace keys only, replayed blocks 0-63. Its known-answer tests (three FIPS 180 SHA-256 vectors, three Random123 Philox4x32-10 vectors, a 64-by-64-bit product, the multiply-high rejection threshold, popcount and coordinate-schema injectivity) all passed. It checked 64 blocks, 512 path records, 131,072 update records and 16,777,216 audit rows, each equal to the expected count, and reported status PASS, zero mismatches and no first discrepancy. The receipt states that it reports no estimate and is not a scientific result.

Record audit. The Python tool MMEM-AUDIT-RECORDS-1.2.0 (source SHA-256 8e551ab55cbe5dc08b94b3fa6152da5f282993d81c00b383e7c8bf43f9d66d7b) generated no random draws. It checked 41,600 block records (all with both SHAM identities), 332,800 path records, 85,196,800 update records, 512 reconstructed audit-path hashes, 16,777,216 audit rows, 97 manifest files, 22 estimates and one decision file, each equal to the expected count, and reported status PASS and zero mismatches. Its self-checks, including the six frozen half-widths, all passed. The records it compared include the estimate file with SHA-256 83828a49936bb8364813e2634f11cb28c4de2212b4b162d96e616ddbc7b4f8c0 and the decision file with SHA-256 5b3aea3c18c4530b2be9f29d5c64fdd8173b6b323823bf9cc3361bd4e7010fc0.

Neither tool imports producer code. Both are internal reproducibility checks performed within the project with AI assistance; neither is external peer review or an independent scientific replication. Producer and audit tools were both written with AI assistance from the same specification; the audits detect implementation divergence, not a shared misreading of the specification.

S14. Exact claim limits

The result concerns selection on use of a supplied private cache under one prospectively aligned recurrent law and one fixed opportunity cost; meaningful population benefit was not demonstrated. Specifically:

- **Supported:** under the HALF law and the one-proposal opportunity cost, the late-window start-averaged frequency of the mutation-generated memory-use allele exceeded both its exact neutral expectation and its ZERO-law frequency by more than one expected individual (95% familywise), and both late-window start contrasts passed the prespecified finite-horizon $\pm 1/32$ start-state gate, that is, they were bounded within one expected individual (95% familywise) over the 256-update horizon.
- **Bounded:** the recurrent-law ACTIVE minus SHAM accuracy contrast is below one correct bit per survivor (95% familywise within the secondary family); its sign is not resolved.
- **Unresolved:** the recurrence interaction in accuracy.

- **Descriptive only:** all 16 cell means, the ZERO allele deficit, the ZERO adverse accuracy contrast, and all diagnostics.
- **Not supported:** origin of memory architecture or capacity; separate selection on cache retention versus retrieval; general start-independence; rare-mutant invasion; fixation probability; stationarity; equilibrium; evolutionary stability; arbitrary recurrence or any other lag, copy probability or target law; any other or general cost scale; population benefit; biological, ecological or neural extrapolation; priority or global novelty; and replication of earlier internal private-memory studies.

S15. Provenance of quoted values (Study 001)

Every number in the manuscript and this supplement that is registered in provenance/QUOTED_STATISTICS.json is checked by scripts/check_statistics.py against the saved records below (Study 001) and in Section S26 (Study 002), which also recomputes the six interval endpoints, the 16 descriptive identities, the decision labels, the scope counts and the audit mismatch counts. The SHA-256 values are those authenticated in the acceptance record of 2026-10-01.

Source	SHA-256
Decision record	5b3aea3c18c4530b2be9f29d5c64fdd8173b6b323823bf9cc3361bd4e7010fc0
22 estimate records	83828a49936bb8364813e2634f11cb28c4de2212b4b162d96e616ddbc7b4f8c0
Descriptive diagnostics	51554620a5e302416edf8de8a073c6962abbc2c8967da80b91b83e4c3d5b685c
Replay receipt	9b40322d0d09b90f0a6862533c1efdc7aff21bd699a882b9b9555c5f7c4b4857
Record-verification receipt	afd96dc0a5b190401c035fc66c8af8fd4d4603e7185d05bff73fb05e698cff5a
Production inventory audit	622fa9e01a40c44ba545926e54b06e102d20b7fcf9e9c8e645ddaf9f7212e09c
Acceptance record	2eb3aaf5d754f4570a75522aa9e1f38358715c06c674d5a267732a1357926e6f

The first six values are authenticated in the acceptance record; the acceptance-record SHA-256 itself is recomputed and enforced by the statistics-checking script.

Prospective timing records (all times UTC, 2026-10-01). These establish design provenance only and add or modify no scientific outcome.

Record	Time	SHA-256
Frozen specification, design/PHASE2-MUTABLE-MEMORY-001-REV1.md (revision 1)	not separately timestamped; subject of the 01:55:25 review	c17a3a1e9ac9cf2260f6743eaf08e4c2808080a767af16d2f9e9d2b9ad3e821d
Terminal design review, non-implementing AI reviewer CLAUDE-003, verdict GO	01:55:25	787cf32a20312f2682c761243f3dec04c5eb8036d14d6efc444e9830db37960a

Record	Time	SHA-256
Terminal design-review receipt	01:55:25	6277757f9ab3dfa611515c161776c17b561738eba6fa478ae0f8634e13c03176
Production execution authority	12:53:04	ee3cb68bed5d212e6ea10ece542e2e90c32cb9cea07d85fceda10fc56b375b1d
Sole production submission, job 53530597 (no automatic retry, no parameter changes)	12:55:19	not supplied
Acceptance record	13:33:13	2eb3aaf5d754f4570a75522aa9e1f38358715c06c674d5a267732a1357926e6f

Part B. Study 002 (PHASE2-PERFORMANCE-CONVERSION-002)

S16. Status, question and relation to Study 001

Study 002 is a post-v1.0 prospective extension specified in PHASE2-PERFORMANCE-CONVERSION-002 revision 1 (SHA-256 58ee414d753fab042e6cccee1a4e6b95dd7f167633b4f33c1491687d9afcb895). Its frozen question was: under the single aligned HALF recurrence law, does directional information in the supplied one-record cache produce a late-window population-accuracy gain exceeding one correct bit when compared with a control that preserves the one-proposal displacement and the cache-parent distance but scrambles cache direction at recurrent updates? It is a within-law operator intervention. It does not test the origin of memory capacity, a per-use value of information, arbitrary recurrence, equilibrium, invasion, biological memory or generality across model classes.

The accepted primary decision is **BOUNDED BELOW THE POSITIVE ONE-BIT SCALE** (rule 5; interval wholly inside $(-1/32, +1/32)$). The secondary classifications are **MEANINGFUL POSITIVE AT THE ONE-BIT SCALE** for E_INFO and E_NONINFO, and **BOUNDED BELOW THE POSITIVE ONE-BIT SCALE** for B_INFO and B_NONINFO. The complete scope is 41,600 new independent paired blocks, six cells per block, 249,600 paths, 63,897,600 path-updates, 8,178,892,800 objective queries and 19 saved estimate records.

Relation to Study 001. All operators other than the policy probe are those of Sections S2-S6. The Study 002 producer extends the authenticated Study 001 producer; the specification required unchanged operators to remain behaviorally and fixture-identical. The cohort, namespace and every random tape are new, and no Study 001 genotype, target, trajectory, block or outcome was reused. The INFO probe rule equals the Study 001 ACTIVE rule. The 22 Study 001 records and the 19 Study 002 records are separate accepted estimate sets, 41 saved estimate records in total, with separate familywise controls; they are not pooled and are not independent replications of each other.

S17. Arms, shared state and full event order

Fixed state is as in Section S2. Initialization (update 0) for each block: 32 independent uniform 32-bit genotypes shared by all six cells; all caches invalid; every label F in ALL_F cells and every label M in ALL_M cells.

At each update $t = 1, \dots, 256$:

1. Generate the innovation I_t and recurrence bit R_t , and set the HALF target: $T_1 = I_1$, $T_2 = I_2$, and for $t \geq 3$, $T_t = T_{(t-2)}$ if $R_t = 1$ and $T_t = I_t$ otherwise. Generate the shared permutation $\pi_i(b,t)$ (Section S18). All are generated whether or not they are used.
2. For each parent slot $i = 0, \dots, 31$ with genotype x and cache c , construct four candidates: the parent x ; the policy probe (table below); the global scout $x \text{ XOR } \text{scout_mask}$; and the local child, the lowest-mismatch member of the first three candidates (keyed donor tie-break) with each bit flipped independently with probability $1/32$.
3. Evaluate all 128 candidates by Hamming mismatch to T_t , in (family, parent slot) order. Duplicates are evaluated separately.
4. Draw 32 survivors sequentially without replacement with exact integer weights $2^{(32-h)}$ (Section S5). No label, cache, lineage or arm enters the weights.
5. Each survivor inherits its producing parent's policy; each survivor slot then flips $M \leftrightarrow F$ independently with probability $1/32$.
6. Post-mutation F survivors receive an invalid cache; post-mutation M survivors receive the producing parent's pre-update genotype.
7. Record memory-use count, total mismatch, valid probe use, true-cache use, decoy use, probe survivors, policy mutations and the query count, all after step 6.

Policy probe by arm:

Label and cache state at update t	INFO	NONINFO	SHAM
F, or M with invalid cache	$x \text{ XOR } \text{fresh_mask}$	$x \text{ XOR } \text{fresh_mask}$	$x \text{ XOR } \text{fresh_mask}$
valid M, $t < 3$ or $R_t = 0$ (innovation update)	c	c	$x \text{ XOR } \text{fresh_mask}$
valid M, $t \geq 3$ and $R_t = 1$ (recurrent update)	c	$x \text{ XOR } \pi_i(b,t)(c \text{ XOR } x)$	$x \text{ XOR } \text{fresh_mask}$

The experimental operator reads R_t ; the evolving policy cannot. At innovation updates INFO and NONINFO apply the identical rule, so they remain bit-identical until the first recurrent valid-M coordinate, where only the policy-probe genotype may differ before downstream selection. Both arms displace one fresh proposal at every valid-M use.

S18. The shared decoy permutation

Construction. For each block b and update t , $\pi_i(b,t)$ is built from the identity array $[0, \dots, 31]$ by a Fisher-Yates shuffle: for $i = 31, \dots, 1$, draw j exactly uniformly from $[0, i]$ and swap positions i and j . A source bit s maps to destination $\pi[s]$. Each j is drawn by multiply-high (Lemire) rejection from the purpose-separated stream DECOY_PERMUTATION, whose Philox counter is (block, update, entity = i , subindex = retry) and whose unsigned 64-bit draw is $\text{word0 OR } (\text{word1} \ll 32)$; output words 2-3 are unused. Every step is generated for every block and update regardless of branch or arm, and one permutation is shared by all six cells and all 32 parents of that block and update.

Random coordinates. Study 002 uses canonical Random123 Philox4x32-10 with the key and counter construction of Section S7 under the UTF-8 namespace PHASE2-PERFORMANCE-CONVERSION-002|production-r1|<purpose>. The purposes are those of Section S7 (INITIAL_GENOTYPE, TARGET_INNOVATION, TARGET_COPY, FRESH_MASK, SCOUT_MASK, LOCAL_BIT, DONOR_KEY, SURVIVAL_UNIFORM, POLICY_MUTATION) plus DECOY_PERMUTATION. All coordinates are addressable, the implementation aborts on range overflow, and the specification required a purpose-separated collision audit and a constructed exact-permutation frequency fixture before production.

Geometry of the control.

- *Distance matched.* Permuting bit positions preserves Hamming weight, so $\text{popcount}(\text{pi}(c \text{ XOR } x)) = \text{popcount}(c \text{ XOR } x)$ and every NONINFO decoy lies exactly as far from its parent as the cache does, pathwise.
- *One common relabeling.* Because one permutation is shared by all parents of a block and update, overlaps among the displacement masks of different parents are preserved under one common coordinate relabeling.
- *Not preserved.* Other cross-genotype relations are not preserved; in particular, siblings that share one cached genotype do not share one decoy endpoint.
- *Uniform subset law.* For a fixed parent and displacement weight w , the decoy displacement is a uniformly random w -subset of the 32 positions.
- *Retained direction.* Displacements of weight 0 or 32 are unchanged by any permutation and therefore retain direction. Their counts are reported in Section S23.

S19. SHAM identities in Study 002

SHAM never reads labels or caches in genotype construction, so identities N1 and N2 of Section S8 hold: SHAM genotype, mismatch, accuracy and survivor trajectories are bit-identical between starts, and SHAM labels are exact complements after every update, so the start-averaged memory-use frequency is exactly 1/2 in every block. The neutral derivation of Section S8 applies unchanged. Observed: SHAM memory-use frequencies were 0.499678 (21285489/42598400) and 0.500322 (21312911/42598400), which sum exactly to 1, and SHAM accuracy was 0.619009 (1687602263/2726297600) from both starts. The record verifier confirmed both identities in all 41,600 blocks. SHAM is both the neutral allele benchmark for E_NONINFO and the no-memory performance reference for B_INFO and B_NONINFO; as in Study 001, its genotypes and genealogies diverge from INFO and NONINFO at the first valid-M retrieval, so contrasts with SHAM are arm-level comparisons, not path-level counterfactuals.

S20. Estimands and interval families

For block b , let $L_b(\text{arm}, \text{start})$ be the mean memory-use frequency over updates 193-256 and $P_b(\text{arm}, \text{start})$ the corresponding mean population accuracy. With equal-weight start averages $A_b(\text{arm})$ and $P_b(\text{arm})$:

- $\Delta_{P,b} = P_b(\text{INFO}) - P_b(\text{NONINFO})$ (range length 2);
- $D_{\text{INFO},b} = L_b(\text{INFO}, \text{ALL_M}) - L_b(\text{INFO}, \text{ALL_F})$ and $D_{\text{NONINFO},b}$ analogously (range length 2);
- $E_{\text{INFO},b} = A_b(\text{INFO}) - A_b(\text{NONINFO})$ (range length 2);
- $E_{\text{NONINFO},b} = A_b(\text{NONINFO}) - 1/2$ (range length 1);

- $B_INFO,b = P_b(INFO) - P_b(SHAM)$ and $B_NONINFO,b = P_b(NONINFO) - P_b(SHAM)$ (range length 2).

Each estimate is the mean of its block variable over $n = 41,600$ independent blocks and equals the same linear combination of the saved descriptive cell means; the statistics-checking script verifies these identities exactly in rational arithmetic. Δ_P is the total population-accuracy effect of directional cache information beyond the parent and matched displacement length, mediated through evolved policy use: INFO and NONINFO share policy rules but not realized policy states, so Δ_P includes any divergence in policy frequency, cache use, survivor composition and descendants. It is not a per-use value-of-information estimate and does not separate retention from retrieval.

Intervals use the Hoeffding-Bonferroni construction of Section S9, $h = R \sqrt{\ln(2/\alpha_each)/(2n)}$, with exact rational estimates and high-precision half-widths. No empirical variance, normal approximation or pilot is used.

Family	Estimands	alpha_each	Range length	Half-width
Primary	Δ_P , D_INFO , $D_NONINFO$	0.05/3	2	0.01517128
Secondary allele	E_INFO	0.025	2	0.01451463
Secondary allele	$E_NONINFO$	0.025	1	0.00725731
Secondary performance	B_INFO , $B_NONINFO$	0.025	2	0.01451463

The primary half-width lies below $\Delta/2 = 0.015625$, so the gate can be passed only if a start contrast is estimated near zero. Each family has 95% familywise coverage; coverage is not controlled jointly across the three Study 002 families or with the Study 001 families.

S21. Frozen decision order and its application

Primary rules, applied in order with $\delta = 1/32 = 0.03125$:

1. Any deterministic identity, coordinate, query-count, saved-output or independent-audit failure: **INVALID**. Not triggered.
2. Either gate interval not wholly within $(-\delta, +\delta)$: **START-DEPENDENT; PRIMARY UNRESOLVED**. Not triggered: D_INFO $[-0.015042, 0.015301]$ and $D_NONINFO$ $[-0.016439, 0.013903]$ both lie inside.
3. Lower bound of Δ_P above δ : meaningful positive directional-information effect relative to NONINFO. Not triggered: the lower bound is -0.006521 .
4. Upper bound of Δ_P below $-\delta$: meaningful adverse directional-information effect relative to NONINFO. Not triggered.
5. Upper bound of Δ_P at or below δ : **BOUNDED BELOW THE POSITIVE ONE-BIT SCALE**, with a separate statement of whether the interval is wholly inside $(-\delta, +\delta)$. Triggered: the upper bound is 0.023821 , and the interval $[-0.006521, 0.023821]$ lies wholly inside $(-\delta, +\delta)$.
6. Otherwise: UNRESOLVED. Not reached.

The decision record lists decision rule 5 as applied and records that the bounded interval lies wholly inside $(-\delta, +\delta)$. Each secondary estimand was classified, in order, as meaningful positive if its lower bound exceeds δ , meaningful adverse if its upper bound is below $-\delta$, bounded below the positive one-bit scale if its upper bound is at or below δ , and unresolved otherwise. E_INFO (lower bound 0.169688) and $E_NONINFO$ (lower bound 0.118825) are **MEANINGFUL POSITIVE AT THE ONE-BIT SCALE**; B_INFO (upper bound 0.023896) and $B_NONINFO$ (upper bound 0.015246) are **BOUNDED BELOW THE POSITIVE ONE-BIT SCALE**.

Secondary classifications cannot alter the primary decision, and a positive primary would not have established benefit over no-memory use unless the lower bound of B_INFO also exceeded delta.

Prespecified interpretive branches:

Pattern	Frozen interpretation	Applied
Positive primary and meaningful positive B_INFO	One-bit benefit relative to both the matched decoy and SHAM in this fixed model	No
Positive primary without meaningful B_INFO	Directional information offsets some displacement cost; no demonstrated one-bit absolute benefit	No
Bounded or adverse primary with meaningful positive E_INFO	Directional information selects for policy use without converting into one-bit population performance, sharpening the selection-performance separation.	Yes
Bounded or adverse primary without meaningful E_INFO	The information intervention does not explain the accepted enrichment	No

All branches were to be retained as reported. No parameter, lag, probability, cost, horizon or cohort follow-up follows any branch.

S22. All 19 saved estimate records

Inferential records carry frozen bounds; descriptive records do not. Exact values are the saved rational estimates; decimal values are rounded to six places.

#	Record	Family	Exact	Estimate	Interval
1	Delta_P	Primary	5895597/681574400	0.008650	[-0.006521, 0.023821]
2	D_INFO	Primary	2757/21299200	0.000129	[-0.015042, 0.015301]
3	D_NONINFO	Primary	-54019/42598400	-0.001268	[-0.016439, 0.013903]
4	E_INFO	Secondary allele	1569351/8519680	0.184203	[0.169688, 0.198718]
5	E_NONINFO	Secondary allele	10741847/85196800	0.126083	[0.118825, 0.133340]
6	B_INFO	Secondary performance	25576047/2726297600	0.009381	[-0.005133, 0.023896]
7	B_NONINFO	Secondary performance	1993659/272629760	0.000731	[-0.013783, 0.015246]
8	Frequency, INFO, ALL_F	Descriptive	69028243/85196800	0.810221	descriptive
9	Frequency, INFO, ALL_M	Descriptive	69039271/85196800	0.810351	descriptive

#	Record	Family	Exact	Estimate	Interval
10	Frequency, NONINFO, ALL_F	Descriptive	26697133/42598400	0.626717	descriptive
11	Frequency, NONINFO, ALL_M	Descriptive	13321557/21299200	0.625449	descriptive
12	Frequency, SHAM, ALL_F	Descriptive	21285489/42598400	0.499678	descriptive
13	Frequency, SHAM, ALL_M	Descriptive	21312911/42598400	0.500322	descriptive
14	Accuracy, INFO, ALL_F	Descriptive	856541369/1363148800	0.628355	descriptive
15	Accuracy, INFO, ALL_M	Descriptive	856636941/1363148800	0.628425	descriptive
16	Accuracy, NONINFO, ALL_F	Descriptive	1689605501/2726297600	0.619744	descriptive
17	Accuracy, NONINFO, ALL_M	Descriptive	1689586343/2726297600	0.619737	descriptive
18	Accuracy, SHAM, ALL_F	Descriptive	1687602263/2726297600	0.619009	descriptive
19	Accuracy, SHAM, ALL_M	Descriptive	1687602263/2726297600	0.619009	descriptive

Saved-precision values of the primary estimand and the secondary contrasts:

Record	Estimate	Lower	Upper
Delta_P	0.00864996836735652	-0.006521316095375593	0.023821252830088634
E_INFO	0.18420304518479567	0.16968841954593594	0.19871767082365541
E_NONINFO	0.12608275193434495	0.11882543911491509	0.13334006475377482
B_INFO	0.009381238130422739	-0.005133387508436994	0.023895863769282472
B_NONINFO	0.0007312697630662184	-0.013783355875793514	0.015245895401925951

Derived start averages used in the main text: memory-use frequency 0.810286 for INFO and 0.626083 for NONINFO (exactly 1/2 for SHAM); accuracy 0.628390 for INFO, 0.619740 for NONINFO and 0.619009 for SHAM. In percentage points, Delta_P is +0.865 (-0.652 to +2.382), E_INFO 18.420 (16.969 to 19.872), E_NONINFO 12.608 (11.883 to 13.334), B_INFO +0.938 (-0.513 to +2.390) and B_NONINFO +0.073 (-1.378 to +1.525).

S23. Descriptive diagnostics

Totals over all blocks and updates, from the Study 002 descriptive diagnostics record. These counts support interpretation only and carry no inferential bounds. Valid probe use counts policy probes built by valid-M carriers in INFO or NONINFO; in NONINFO it is the sum of true-cache uses (innovation updates) and decoy uses (recurrent updates).

Cell	F -> M	M -> F	Valid probe uses	Valid probe survivors	Valid-M parents at displacement weight 0
INFO, ALL_F	2,222,248	8,425,691	262,335,134	76,070,068	83,507,588
INFO, ALL_M	1,745,959	8,901,980	276,680,738	79,832,243	87,969,874
NONINFO, ALL_F	4,298,024	6,349,915	200,347,494	47,690,642	84,354,781
NONINFO, ALL_M	3,687,279	6,960,660	218,682,953	51,883,813	91,669,182
SHAM, ALL_F	5,653,523	4,994,416	0	0	50,037,470
SHAM, ALL_M	4,994,416	5,653,523	0	0	56,000,609

Cell	True-cache uses	True-cache survivors	Decoy uses	Decoy survivors	Decoys of weight 0	Decoys identical to cache
INFO, ALL_F	262,335,134	76,070,068	0	0	0	0
INFO, ALL_M	276,680,738	79,832,243	0	0	0	0
NONINFO, ALL_F	100,189,242	19,329,879	100,158,252	28,360,763	42,175,734	42,478,033
NONINFO, ALL_M	109,946,211	21,266,901	108,736,742	30,616,912	45,696,359	46,024,351

The SHAM mutation counts are exchanged between starts, as label complementarity (N2) implies. Every cell contains 5,284,414 recurrent block-updates. No displacement of weight 32 occurred in any arm, and no survival or permutation retry occurred. Weight-0 displacements, where the cache equals the parent and the decoy necessarily equals the cache, made up 42.1% (ALL_F) and 42.0% (ALL_M) of NONINFO decoy applications; these retain direction trivially. Decoys identical to the cache slightly outnumber weight-0 decoys because a permutation can also leave a nonzero displacement unchanged. In every one of the 41,600 blocks, NONINFO paths diverged from INFO paths from both starts. The diagnostics record (SHA-256 in Section S26) also archives fixation, extinction and late-window occupancy counters; they are not interpreted here. It lists 8,178,892,800 total objective queries.

S24. Execution history, fixtures and independent audit

Failed original production submission (infrastructure only). The original Study 002 production submission did not run the model. It stopped at command-line validation because the required --design-go argument was missing, and exited with status 64. It created no production output, generated no random number, executed no model and produced no outcome; no value from it exists or was used. The specification states that a failed or uncertain submission is never duplicated. The authenticated failure record has SHA-256 289bf3188286476698136d499c12eb3a8c9e4e6fc5cde18fcb15718395f4afaa and was accepted at 06:28:45 UTC on 2026-10-02. A prospective recovery addendum (SHA-256 db12c0143b6ba2c0b8cd3771ad18b9afe12feb069fe2a97d15227f4fad836e83) was issued at 06:29:59 before the corrected run. The corrected run used the frozen production namespace (production-r1) and the frozen scientific counts and wrote to a separate attempt-2 production directory.

Jobs. Production job 53553084 (run manifest SHA-256 b7aceaf6638e8f06246c8fb9b182c280a8ed98c4444d93a4a60decdbafb80136; runner receipt 6ea7f69901c17005a017cf134fd8804a7bdd0c1f4b43de53f9e28e0044de8087); frozen analyzer job 53553256 (runner receipt c6bb16016b0133bea52dc3a8e4db5b8fe13189bdd0c79a231181c4669b321570); separately implemented audit job 53553294 (runner receipt 32e3ffff29b3de9f20158af58a67c2c130cb7456ec8c7287f67354897928a8ed). The specification allowed one production allocation of at most 32 CPUs, 64 GiB RAM and six hours.

Deterministic fixtures required before production (none contains a sampled production outcome). (1) Study 001 candidate, survival, mutation and cache hand traces bit-identical when the new arm is absent; (2) Fisher-Yates and rejection fixtures with forced retries and exact known permutations; (3) INFO and NONINFO bit-identical through every innovation update while their states are identical; (4) at the first recurrent valid-M coordinate, only the policy-probe genotype may differ before downstream selection; (5) decoy distance equal to cache distance for all 33 possible weights, with weights 0 and 32 unchanged and reported; (6) preservation of Hamming weights and pairwise overlaps of displacement masks under the shared permutation; (7) the uniform subset law, verified analytically and by exhaustive small-bit analogues; (8) SHAM N1 and N2 for full 256-update paths from both starts and a mixed-label diagnostic; (9) unconditional generation of all random purposes, with coordinate-collision and branch-invariance audits; (10) bit-identical regeneration of a non-audit path from its block identifier and source and configuration hashes.

Replay audit. The C++ tool PCONV-AUDIT-REPLAY-1.0.0 (executable SHA-256 48fff49ee3c704adb0ea46f57823fa7830e7f1fa5ec9d7575dde9aca730e4ac3), run in production mode with production-namespace keys only, replayed blocks 0-63. Its known-answer tests (three FIPS 180 SHA-256 vectors, three Random123 Philox4x32-10 vectors, the key-word layout and Study 002 key texts including DECOY_PERMUTATION, a 64-by-64-bit product, the survival and Fisher-Yates rejection thresholds, popcount, published FNV-1a vectors, coordinate-schema ranges and donor injectivity, identity and rotation Fisher-Yates cases with source-to-destination orientation, a forced rejection retry, exhaustive 4- and 5-position analogues of the uniform subset law, decoy distance matching at weights 0 to 32 with retained weights 0 and 32 and preserved overlaps, the arm-specific probe rule and the HALF law) all passed. It checked 64 blocks, 384 path records, 98,304 update records, 12,582,912 audit rows and 16,384 permutation records, each equal to the expected count, and reported status PASS, zero mismatches and no first discrepancy. The receipt states that it reports no estimate and is not a scientific result.

Record audit. The Python tool PCONV-AUDIT-RECORDS-1.0.0 (source SHA-256 e700ce2621def54db2b2f14fb077dced6cff0e57eb6a3439ac2cfe870e1b5ad9) generated no random draws. It checked 41,600 block records (all with both SHAM identities), 249,600 path records, 63,897,600 update records, 63,897,600 path-updates with 128 queries, 384 reconstructed audit-path hashes, 12,582,912 audit rows, 16,384 permutation records, 98 manifest files, 19 estimates and one decision file, each equal to the expected count, and reported status PASS and zero mismatches. Its self-checks, including the seven frozen half-widths and a Fisher-Yates replay check that detects a wrongly accepted draw, all passed. The records it compared include the estimate file with SHA-256 5efa213cdfef985a20fd7bbc65fb36dd505c568670ad350b3223c2139eb0b68d and the decision file with SHA-256 3f28b82d0cb40357a1068c464d1c5158c36ba8fbfe775026ce22c70113b3b84a.

The specification required a reviewer that did not implement the producer to reconstruct the estimates and replay the audit paths without importing producer code. Both audits are nevertheless internal reproducibility checks performed within the project; neither is external peer review or an independent scientific replication, and they detect implementation divergence, not a shared misreading of the specification.

S25. Exact claim limits for Study 002 and the two-study result

The result concerns the total population-accuracy and allele-frequency effects of directional information in a supplied one-record cache under one HALF law aligned to the cache delay, one fixed one-proposal opportunity cost, one mutation rate, one finite horizon and one model class. Specifically:

- **Supported:** relative to the recurrence-gated, parent-distance-matched NONINFO control, directional cache information raised the late-window start-averaged memory-use frequency by more than one expected individual (95% familywise), and NONINFO itself exceeded exact neutrality by more than one expected individual; both Study 002 start contrasts passed the prespecified finite-horizon $\pm 1/32$ gate.
- **Bounded:** the total population-accuracy effect of directional information relative to NONINFO (Δ_P) lies below one correct bit per survivor, wholly inside $\pm 1/32$; B_INFO and B_NONINFO lie below one correct bit; no sign is resolved.
- **Descriptive only:** all 12 cell means, all diagnostics, and the close agreement between the Study 002 INFO and Study 001 ACTIVE-HALF cell means.
- **Joint result:** recurrence selected a costly memory-use policy (Study 001), and directional cache information strongly amplified that selection (Study 002), without a resolved one-bit population-performance gain in either study.
- **Not supported:** a per-use value-of-information or per-use causal effect; separate selection on retention versus retrieval; spontaneous origin of memory architecture or capacity; general population utility; any other lag, law, cost, mutation rate, horizon or model class; equilibrium, invasion, stability or fixation; biological, ecological or neural relevance; priority or global novelty; independent replication of Study 001 or independent model-class reproduction.

The control matches parent distance and one common coordinate relabeling but not every cross-genotype relation, and weight-0 displacements retain direction. No grid, cost sweep, new cohort, lag variant or numerical extension follows from either study.

S26. Provenance of Study 002 quoted values

Every registered Study 002 value in the manuscript and this supplement is checked by scripts/check_statistics.py against the saved records below, which also recomputes the seven interval endpoints, the exact linear identities, the decision and classification labels, the SHAM identities, the scope counts, the diagnostic consistency relations and the audit mismatch counts. The checker requires both the 22 Study 001 records and the 19 Study 002 records and authenticates each set against its own acceptance record.

Source	SHA-256
Decision record	3f28b82d0cb40357a1068c464d1c5158c36ba8fbfe775026ce22c70113b3b84a
19 estimate records	5efa213cdfefb985a20fd7bbc65fb36dd505c568670ad350b3223c2139eb0b68d
Descriptive diagnostics	304843eb5d4a38761a6ce371c7cc125260041dfd910b29d5ba3cc377a6ac3553
Replay receipt	cbee79527876745e8918576f73a5d302377e4463d959318ca2f5feffa66416d4
Record-verification receipt	09f5dccf4921c8d842ef60fb31aca712bbb582e87a2011c76c02efb658539197
Production run manifest	b7aceaf6638e8f06246c8fb9b182c280a8ed98c4444d93a4a60decdbafb80136
Acceptance record (accepted 2026-10-02 07:10:51 UTC)	fc7777239cd4442c35c01d6261fd71e6fbf7e7f6fd6e11ddc5d24876ce2f4a2
Frozen specification, revision 1	58ee414d753fab042e6cccee1a4e6b95dd7f167633b4f33c1491687d9afcb895
Terminal design review (DESIGN_GO)	76b0c6172e61f60befecf183e83d9e43ea50c1b6bc1e93299668b76f55be60f
Design-GO record (frozen 2026-10-01 18:59:00 UTC)	f4a6180109e0d212cbe784a7722a64576e5e7cb0fbfd4b23c757bb5e7ab84d6a
Original production authority (issued 2026-10-02 06:02:41 UTC)	31c2100dd73de0090ded3059525aacd139565917086d2c19880b01e319c98992
Attempt-1 failure record (accepted 2026-10-02 06:28:45 UTC)	289bf3188286476698136d499c12eb3a8c9e4e6fc5cde18fcb15718395f4afaa
Prospective recovery addendum (issued 2026-10-02 06:29:59 UTC)	db12c0143b6ba2c0b8cd3771ad18b9afe12feb069fe2a97d15227f4fad836e83
Recovery execution authority (issued 2026-10-02 06:32:55 UTC)	c4cb2e4c162df92a761b2fff391b6e81e000055db88d7cbaeb954b0750b4e085

The first six values are authenticated in the Study 002 acceptance record; the acceptance-record SHA-256 is recomputed and enforced by the statistics-checking script; the specification hash is quoted from the source-binding record and is not recomputed by the checker. The remaining provenance records establish prospective order and preserve the disclosed infrastructure-only failure; they add or modify no scientific outcome.